Experimental Design & Scope of Inference

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Observational studies vs. experiments**

A researcher records the exercise habits and cholesterol levels of 500 people without intervening in their lifestyle. Identify whether this is an observational study or an experiment, and explain what type of conclusion (association or causation) it can support.

This is an observational study, since the researcher only records existing behavior without imposing any treatment. It can only support conclusions about association between exercise and cholesterol, not causation, since confounding variables (diet, genetics, age) could explain the relationship instead.

2 **[Foundational]** • **Target: Observational studies vs. experiments**

A researcher randomly assigns 200 volunteers to either an exercise program or a no-exercise control group, then measures cholesterol after 6 months. Identify this design, and explain why it can support a causal conclusion.

This is an experiment, since the researcher actively assigns a treatment (exercise program). Random assignment balances out confounding variables between the two groups on average, so any significant difference in cholesterol can be attributed to the exercise program itself, supporting a causal conclusion.

3 **[Foundational]** • **Target: The three principles of experimental design (RRCB)**

State the three key principles of a well-designed experiment (sometimes remembered as RRC: Random assignment, Replication, Control of confounding variables), and briefly explain the purpose of each.

Random assignment: randomly assigning subjects to treatment groups balances out lurking variables between groups. Replication: using enough subjects per treatment group reduces the role of chance variation, making real treatment effects easier to detect. Control: keeping other variables constant across groups (e.g. a comparison/control group, placebo, or blinding) isolates the effect of the treatment being studied.

4 **[Foundational]** • **Target: The three principles of experimental design (RRCB)**

A study testing a new fertilizer randomly assigns 60 plants to fertilizer or no-fertilizer groups, but only uses 3 plants per group. Identify the design flaw, and explain the consequence.

The flaw is insufficient replication: with only 3 plants per group, natural plant-to-plant variation could easily produce an apparent difference between groups purely by chance, making it hard to distinguish a real fertilizer effect from random variability.

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5 [Foundational] • Target: Blocking and matched pairs designs

A study on a new medication expects men and women to respond differently. Describe how a randomized block design would be set up, using sex as the blocking variable.

Separate the subjects into two blocks by sex (men and women), then randomly assign subjects to treatment or control within each block separately. This ensures the medication is compared fairly within each sex group, removing sex as a source of confounding variation between the treatment and control groups.

6 [Foundational] • Target: Blocking and matched pairs designs

A researcher wants to compare two brands of running shoes on the same 20 runners, having each runner test both brands (in random order) and comparing their times. Identify this design.

This is a matched pairs design (a special case of blocking where each subject serves as their own block), since each runner is compared to themselves under both treatments rather than to a different group of runners.

7 [Foundational] • Target: Placebo effect and blinding

Explain what the placebo effect is, and describe how a single-blind experiment design controls for it.

The placebo effect is an improvement in a subject's condition simply because they believe they are receiving a treatment, even if the treatment has no active effect. A single-blind design controls for this by giving the control group a placebo that looks identical to the real treatment, so subjects do not know which group they are in, ensuring the psychological effect of expecting treatment applies equally to both groups.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Placebo effect and blinding

Explain the difference between a single-blind and a double-blind experiment.

In a single-blind experiment, only the subjects are unaware of which treatment they are receiving. In a double-blind experiment, neither the subjects nor the people evaluating or administering the treatment (who might otherwise unconsciously bias their assessment) know which group is which, further reducing bias.

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9 **[Intermediate]** • **Target: Scope of inference**

A study uses a simple random sample from the population but is purely observational (no random assignment of treatment). State what type of conclusion (about the population, about causation, both, or neither) is justified, and explain why.

Random sampling (without random assignment) justifies generalizing results to the population the sample was drawn from, but since there was no random assignment of a treatment, causal conclusions are not justified — only an association can be claimed for that population.

10 **[Intermediate]** • **Target: Scope of inference**

A study randomly assigns treatments to a group of volunteers who were not randomly selected from any larger population. State what type of conclusion is justified.

Random assignment (without random sampling) justifies a causal conclusion about the effect of the treatment on the specific subjects studied, but since the volunteers were not randomly selected from a broader population, the results cannot be safely generalized beyond the study's own subjects.

11 **[Intermediate]** • **Target: Putting it together: evaluating a study design**

A pharmaceutical company recruits 300 volunteers (not randomly selected from the general population) and randomly assigns them to receive either a new drug or a placebo, in a double-blind design. Evaluate what conclusions can and cannot be drawn from this study.

Because subjects were randomly assigned to drug or placebo, a causal conclusion about the drug's effect (relative to placebo) on these subjects is justified — including that the double-blind design controls for the placebo effect and evaluator bias. However, because the 300 volunteers were not randomly selected from the general population, the results cannot be safely generalized to the population at large; the study only supports causal claims about the volunteers themselves (or, with caution, people similar to th.

12 **[Intermediate]** • **Target: Confounding variables**

An observational study finds that people who drink more coffee have a higher rate of heart disease. Explain how smoking could act as a confounding variable in this relationship.

If people who drink a lot of coffee are also more likely to smoke, and smoking independently increases the risk of heart disease, then smoking is a confounding variable: it is associated with both coffee drinking and heart disease, making it impossible to tell from this observational data alone whether coffee itself, smoking, or some combination is responsible for the observed relationship.

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13 [Intermediate] • Target: Confounding variables

Explain how a well-designed randomized experiment would help rule out a variable like smoking as a confounder in the coffee and heart disease example.

In a randomized experiment, subjects would be randomly assigned to a coffee-drinking treatment or a control group. Random assignment tends to balance the two groups on other characteristics, including smoking habits, on average — so any difference in heart disease outcomes between the groups could more confidently be attributed to coffee itself, not to smoking or other confounders that random assignment has balanced out.

14 [Intermediate] • Target: Identifying experimental units and treatments

A researcher tests three different fertilizer amounts (low, medium, high) on tomato plants, measuring final yield. Identify the experimental units, the treatment, and the response variable.

Experimental units: the individual tomato plants. Treatment: the fertilizer amount applied (low, medium, or high) — this is the explanatory variable being manipulated. Response variable: the final yield of each plant, which is measured as the outcome.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A pharmaceutical researcher is testing whether a new sleep aid reduces the time it takes people to fall asleep, and worries that people with severe insomnia might respond very differently from people with mild insomnia. Design a study using a randomized block design that accounts for this concern, describing each step, and explain the specific benefit this design provides over a simple completely randomized design in this scenario.

First, classify each subject into a block based on insomnia severity (e.g. "mild" and "severe"). Then, within each block separately, randomly assign subjects to receive either the new sleep aid or a placebo. This ensures the comparison between sleep aid and placebo happens fairly within each severity group, rather than risking, by chance, that a completely randomized design places most severe-insomnia subjects into just one of the two treatment groups — blocking removes insomnia severity as a source of confounding.

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16 **[Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A company tests a new website design by randomly assigning visitors to see either the old design or the new design, without telling visitors which version is the "experimental" one, and measures the percentage who complete a purchase. Explain why this study can support a causal conclusion about the website redesign, and describe the specific scope (which population) that conclusion applies to.

Because visitors are randomly assigned to see the old or new design, random assignment balances out any other differences between the two groups of visitors on average, so a difference in purchase completion rates can be causally attributed to the website design itself. However, the causal conclusion applies specifically to the population of visitors who came to this particular website during the study period — generalizing to a broader population (e.g. all internet shoppers everywhere) would require that these vis.

17 **[Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 1.01 Measures of Center & Spread**

A university studies whether a new tutoring program improves final exam scores. Volunteers who sign up for tutoring are compared to students who did not sign up, without any random assignment. At the end of the semester, tutored students score noticeably higher on average. Explain why this study cannot establish that the tutoring program caused the improvement, identifying a specific plausible confounding variable.

Since students self-selected into the tutoring group rather than being randomly assigned, the tutored and non-tutored groups may differ in other important ways besides tutoring itself — for example, students motivated enough to seek out extra tutoring may also be more diligent, better organized, or more academically engaged in general (a plausible confounding variable: student motivation/effort). This makes it impossible to separate the effect of tutoring itself from the effect of the underlying motivation that led.

18 **[Challenging] • Target: The three principles of experimental design (RRCB) • Cross-Lesson: 1.02 Z-Scores & Linear Transformations**

An agronomist wants to test whether a new organic fertilizer increases tomato yield compared to a standard commercial fertilizer. Explain how the agronomist must incorporate the four principles of experimental design: Comparison, Random Assignment, Control, and Replication.

1. Comparison: Use two treatment groups (organic fertilizer vs. standard fertilizer control) grown concurrently under the same environmental conditions. 2. Random Assignment: Assign tomato plants to fertilizer treatments using a random number generator to balance unmeasured variables (e.g. baseline soil nutrient variations, microclimates). 3. Control: Keep all extraneous variables identical for both groups: sunlight exposure, watering schedule, soil volume, and pest control. 4. Replication: Apply each fertilizer to.

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19 [Challenging] • Target: Blocking and matched pairs designs • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A health study surveys 5,000 adults and finds that people who sleep 8 hours per night have significantly lower blood pressure than those who sleep 5 hours. (a) Can researchers conclude that increasing sleep causes a reduction in blood pressure? (b) Delineate the scope of inference using the 2×2 selection/assignment matrix.

(a) Causal Inference: No, cause-and-effect cannot be established because this is an observational study where sleep duration was not randomly assigned. Plausible confounding variables (e.g. job stress, diet, exercise) are associated with sleep duration and independently affect blood pressure. (b) Scope of Inference:Were individuals randomly selected from the adult population? Yes ⇒ Inferences can be generalized to the broader adult population.Were treatments randomly assigned? No ⇒ Inference.

20 [Challenging] • Target: Placebo effect and blinding • Cross-Lesson: 2.01 Sampling Methods & Bias

Level 3 · AdvancedAn investigator conducts a hypothesis test at significance level $\alpha = 0.05$. Explain two distinct ways the investigator could increase the statistical power of the test without changing the true effect size.

1. Increase the sample size n: A larger sample size reduces the standard error of the sample statistic ($\sigma/\sqrt{n}$), making the sampling distributions of H_0 and H_a narrower, which increases the probability of rejecting a false null hypothesis (power = $1 - \beta$). 2. Increase the significance level α (e.g., from 0.05 to 0.10): A larger α moves the critical rejection boundary closer to the center of H_0, making it easier to reject H_0 and thereby increasing power (at the expense o.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Observational studies vs. experiments

A researcher randomly assigns 467 volunteers to receive a daily supplement or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Observational study

b. Census

c. **Experiment ✓**

d. Meta-analysis

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

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22 [Foundational] • Target: The three principles of experimental design (RRCB)

A researcher records data on exercise habits and cholesterol levels from 435 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study ✓

b. Experiment

c. Meta-analysis

d. Census

Observational study. The researcher only records existing behavior without imposing any treatment — this is an observational study.

23 [Foundational] • Target: Blocking and matched pairs designs

A researcher randomly assigns 208 volunteers to receive a new exercise program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Observational study

b. Census

c. Experiment ✓

d. Meta-analysis

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

24 [Foundational] • Target: Placebo effect and blinding

A researcher records data on fertilizer use and crop yield from 438 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study ✓

b. Experiment

c. Census

d. Meta-analysis

Observational study. The researcher only records existing behavior without imposing any treatment — this is an observational study.

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25 [Foundational] • Target: Scope of inference

A researcher randomly assigns 259 volunteers to receive a new medication or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Census

b. Meta-analysis

c. Observational study

d. **Experiment ✓**

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

26 [Foundational] • Target: Putting it together: evaluating a study design

A researcher randomly assigns 108 volunteers to receive a meditation app or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. **Experiment ✓**

b. Census

c. Meta-analysis

d. Observational study

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

27 [Foundational] • Target: Confounding variables

A researcher randomly assigns 476 volunteers to receive a tutoring program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Census

b. Meta-analysis

c. Observational study

d. **Experiment ✓**

Experiment. The researcher actively assigns a treatment to subjects using random assignment, rather than merely observing existing behavior — this is an experiment.

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28 [Foundational] • Target: Identifying experimental units and treatments

A researcher records data on medication use and symptom severity from 391 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study ✓

b. Experiment

c. Census

d. Meta-analysis

Observational study. The researcher only records existing behavior without imposing any treatment — this is an observational study.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Observational studies vs. experiments

A researcher records data on meditation habits and stress levels from 301 people without intervening in their lifestyle in any way. Identify the type of study.

a. Experiment

b. Observational study ✓

c. Census

d. Meta-analysis

Observational study. The researcher only records existing behavior without imposing any treatment — this is an observational study.

30 [Intermediate] • Target: The three principles of experimental design (RRCB)

In a study on medication use and symptom severity, a simple random sample of 130 people is selected from the population of interest; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?

a. Causation, but only about the subjects actually studied (not generalizable)

b. Causation, generalizable to the entire population ✓

c. Association only, and only about the subjects actually studied (not generalizable)

d. Association only, generalizable to the entire population

Random sampling was used, and random assignment of treatment was used. Causation, generalizable to the entire population.

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31 [Intermediate] • Target: Blocking and matched pairs designs

In a study on fertilizer use and crop yield, 58 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a new fertilizer or a control condition. What type of conclusion is justified by this design?

a. **Causation, but only about the subjects actually studied (not generalizable) ✓**

b. Association only, generalizable to the entire population

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

Random sampling was NOT used, and random assignment of treatment was used. Causation, but only about the subjects actually studied (not generalizable).

32 [Intermediate] • Target: Placebo effect and blinding

In a study on tutoring attendance and exam performance, a simple random sample of 361 people is selected from the population of interest; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. **Association only, generalizable to the entire population ✓**

c. Causation, but only about the subjects actually studied (not generalizable)

d. Association only, and only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was NOT used. Association only, generalizable to the entire population.

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33 [Intermediate] • Target: Scope of inference

In a study on tutoring attendance and exam performance, 127 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?

a. Association only, generalizable to the entire population

b. Causation, but only about the subjects actually studied (not generalizable)

c. Causation, generalizable to the entire population

d. **Association only, and only about the subjects actually studied (not generalizable) ✓**

Random sampling was NOT used, and random assignment of treatment was NOT used. Association only, and only about the subjects actually studied (not generalizable).

34 [Intermediate] • Target: Putting it together: evaluating a study design

In a study on tutoring attendance and exam performance, a simple random sample of 275 people is selected from the population of interest; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?

a. **Causation, generalizable to the entire population ✓**

b. Association only, generalizable to the entire population

c. Association only, and only about the subjects actually studied (not generalizable)

d. Causation, but only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was used. Causation, generalizable to the entire population.

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35 [Intermediate] • Target: Confounding variables

In a study on tutoring attendance and exam performance, 459 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?

a. **Causation, but only about the subjects actually studied (not generalizable) ✓**

b. Association only, generalizable to the entire population

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

Random sampling was NOT used, and random assignment of treatment was used. Causation, but only about the subjects actually studied (not generalizable).

36 [Intermediate] • Target: Identifying experimental units and treatments

In a study on medication use and symptom severity, 471 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?

a. Association only, and only about the subjects actually studied (not generalizable)

b. **Causation, but only about the subjects actually studied (not generalizable) ✓**

c. Causation, generalizable to the entire population

d. Association only, generalizable to the entire population

Random sampling was NOT used, and random assignment of treatment was used. Causation, but only about the subjects actually studied (not generalizable).

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37 [Intermediate] • Target: Observational studies vs. experiments

In a study on medication use and symptom severity, a simple random sample of 476 people is selected from the population of interest; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. **Association only, generalizable to the entire population ✓**

c. Association only, and only about the subjects actually studied (not generalizable)

d. Causation, but only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was NOT used. Association only, generalizable to the entire population.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The three principles of experimental design (RRCB) • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

In a study on medication use and symptom severity, 165 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. **Association only, and only about the subjects actually studied (not generalizable) ✓**

c. Association only, generalizable to the entire population

d. Causation, but only about the subjects actually studied (not generalizable)

Random sampling was NOT used, and random assignment of treatment was NOT used. Association only, and only about the subjects actually studied (not generalizable).

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39 [Challenging] • Target: Blocking and matched pairs designs • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

In a study on supplement use and energy levels, a simple random sample of 406 people is selected from the population of interest; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?

a. Association only, and only about the subjects actually studied (not generalizable)

b. **Causation, generalizable to the entire population ✓**

c. Association only, generalizable to the entire population

d. Causation, but only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was used. Causation, generalizable to the entire population.

40 [Challenging] • Target: Placebo effect and blinding • Cross-Lesson: 2.01 Sampling Methods & Bias

In a study on supplement use and energy levels, 172 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?

a. **Causation, but only about the subjects actually studied (not generalizable) ✓**

b. Causation, generalizable to the entire population

c. Association only, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

Random sampling was NOT used, and random assignment of treatment was used. Causation, but only about the subjects actually studied (not generalizable).

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41 [Challenging] • Target: Scope of inference • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

In a study on fertilizer use and crop yield, a simple random sample of 192 people is selected from the population of interest; researchers simply record whether each person already uses a new fertilizer or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. **Association only, generalizable to the entire population ✓**

c. Causation, but only about the subjects actually studied (not generalizable)

d. Association only, and only about the subjects actually studied (not generalizable)

Random sampling was used, and random assignment of treatment was NOT used. Association only, generalizable to the entire population.

42 [Challenging] • Target: Putting it together: evaluating a study design • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

In a study testing a new medication, subjects know whether they are receiving the real treatment or a placebo, and so do the researchers evaluating the results. Identify the blinding used.

a. **Not blinded ✓**

b. Double-blind

c. Single-blind

d. Triple-blind

Not blinded. Neither the subjects nor the evaluators are kept unaware of group assignment, so there is no blinding at all.

43 [Challenging] • Target: Confounding variables • Cross-Lesson: 3.03 The Normal Distribution

In a study testing a new fertilizer, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.

a. Not blinded

b. **Single-blind ✓**

c. Triple-blind

d. Double-blind

Single-blind. Only the subjects are unaware of their group assignment, while the evaluators know — this is single-blind.

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44 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In a study testing a new fertilizer, neither the subjects nor the researchers evaluating the results know which subjects received the real treatment versus a placebo. Identify the blinding used.

a. Double-blind ✓

b. Triple-blind

c. Single-blind

d. Not blinded

Double-blind. Both the subjects and the evaluators are unaware of group assignment — this is double-blind.

45 [Challenging] • Target: Observational studies vs. experiments • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

In a study testing a new exercise program, subjects know whether they are receiving the real treatment or a placebo, and so do the researchers evaluating the results. Identify the blinding used.

a. Double-blind

b. Single-blind

c. Not blinded ✓

d. Triple-blind

Not blinded. Neither the subjects nor the evaluators are kept unaware of group assignment, so there is no blinding at all.

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